

Trainings ▾

Preparatory Steps

with Mechanically Actuated Injector

- Remove the air crossover. Refer to Procedure 010-019 in Section 10. (</qs3/pubsys2/xml/en/procedures/20/20-010-019-tr.html>)
- Remove the crankcase breather hose. Refer to Procedure 003-001 in Section 13. (</qs3/pubsys2/xml/en/procedures/20/20-003-001-tr.html>)



with Electronically Actuated Injector

- Disconnect the crankcase breather hose. Refer to Procedure 003-001 in Section 3. (</qs3/pubsys2/xml/en/procedures/20/20-003-001-tr.html>)
- Remove the high-pressure injector supply lines. Refer to Procedure 006-051 in Section 6. (</qs3/pubsys2/xml/en/procedures/20/20-006-051-tr.html>)

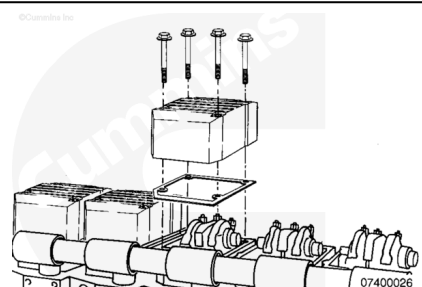


Remove

with Mechanically Actuated Injector

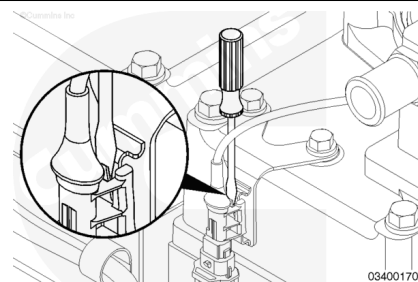
Crankcase breathers can be mounted in any cylinder location. Record the number of breathers and their correct location before removal.

Remove the rocker lever cover and discard the gasket.



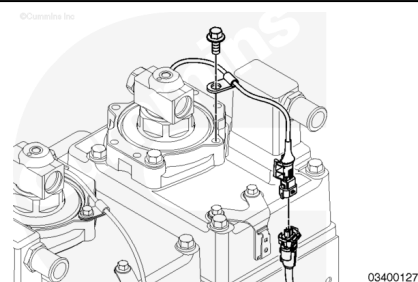
with Electronically Actuated Injector

Remove the wiring harness from the rocker lever cover bracket by using a small flat tip screw driver to release the locking mechanism.



Remove the P-clip securing the injector wiring to the bracket on the injector cover plate.

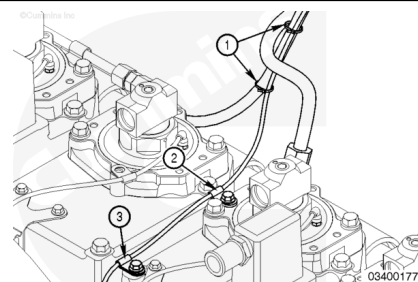
Disconnect the electrical connector to the injector.



For engines with rear turbocharger option:

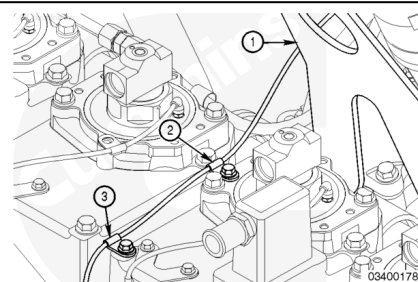
Remove both stainless steel wire ties (1) securing the turbocharger speed sensor wiring to the turbocharger plumbing.

Remove the P-clips securing the turbocharger speed sensor wiring to the injector cover plate (2) and to the rocker lever cover (3).



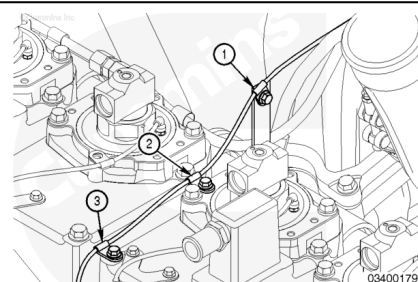
For engines with the front out turbocharger option, with a turbocharger elbow bracket:

Remove the P-clips (1) securing the turbocharger speed sensor wiring to the turbocharger elbow bracket, injector cover plate (2), and to the rocker lever cover (3).



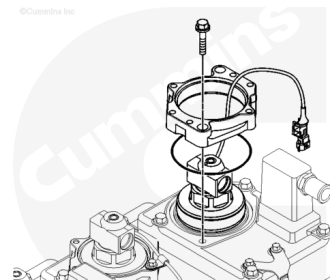
For engines with front out turbocharger option, without a turbocharger elbow bracket:

Remove the P-clips securing the turbocharger speed sensor wiring to the "L" bracket (1), injector cover plate (2) and to the rocker lever cover (3).



Remove the two capscrews securing the injector cover plate to the rocker lever cover. Pry injector cover plate evenly on opposite sides using a heel bar.

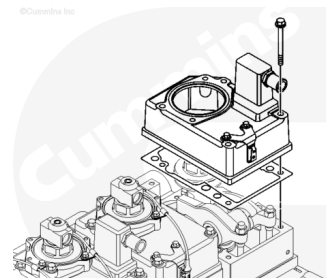
Remove the o-ring seal from the rocker lever cover plate and discard it.



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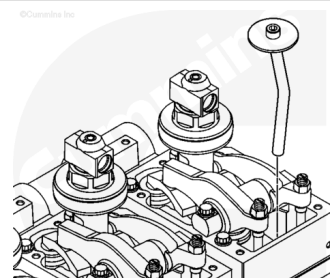
Remove the four capscrews securing the rocker lever cover.

Remove the rocker lever cover, rocker lever cover gasket, and discard the gasket.



03400123

Install the cylinder head protective cover in the hole located between the push tubes to prevent debris from entering the engine while the rocker lever cover is removed.

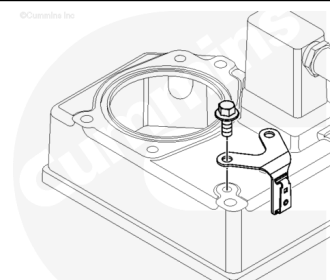


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Disassemble

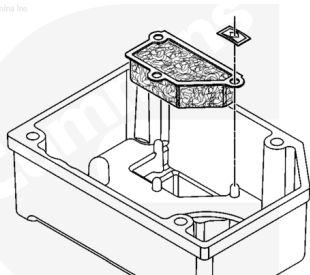
with Electronically Actuated Injector

Remove the two capscrews securing the injector wiring harness bracket to the rocker lever cover plate.



03400124

Remove the breather baffle from the underside of the rocker lever cover by removing the three spring nuts.



03400125

Clean and Inspect for Reuse

with Mechanically Actuated Injector

⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

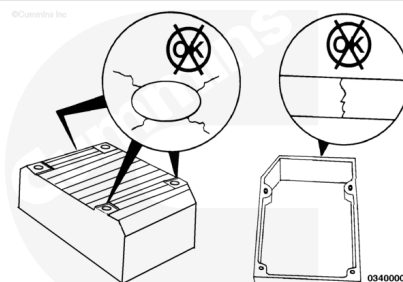
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use solvent to clean the rocker lever covers.

Dry with compressed air.

Inspect the covers for cracks.

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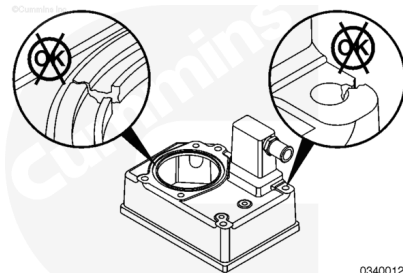
with Electronically Actuated Injector

Inspect all parts for cracks, damage, stripped threads, or damaged holes. Replace any parts that are damaged.

Inspect the injector cover plate dust seal for signs of tears or damage.

Replace if damaged.

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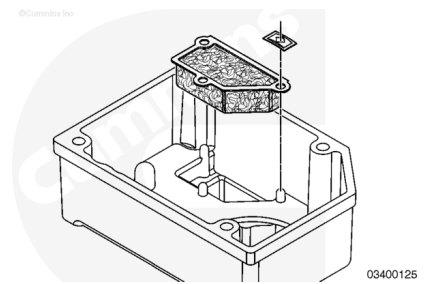


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Assemble

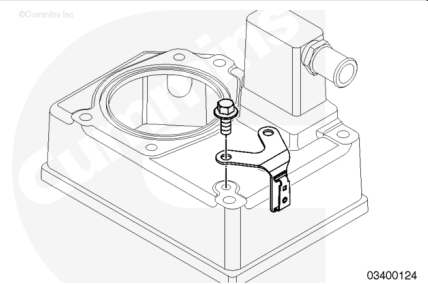
with Electronically Actuated Injector

Install the breather baffle on the three pins on the underside of the rocker lever cover and secure in place with three spring nuts.



Install the injector wiring harness bracket to the rocker lever cover with two capscrews.

Torque Value: 20 n•m [180 in-lb]



Install

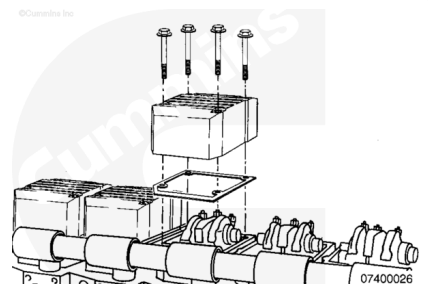
with Mechanically Actuated Injector

⚠ CAUTION ⚠

Do not use gasket cement. Gasket cement will prevent the gasket from sealing properly.

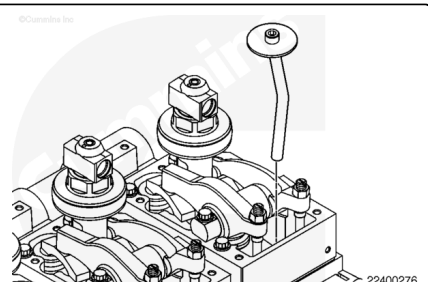
Install the gaskets, rocker lever covers, and capscrews.
Tighten all the capscrews.

Torque Value: 45 n•m [33 ft-lb]



with Electronically Actuated Injector

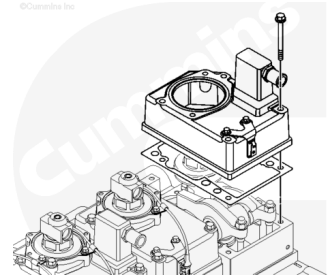
Remove the cylinder head protective cover from the push tube hole.



Install the new rocker lever cover gasket and rocker lever cover.

Secure the rocker lever cover with four capscrews.

Torque Value: 40 n•m [30 ft-lb]

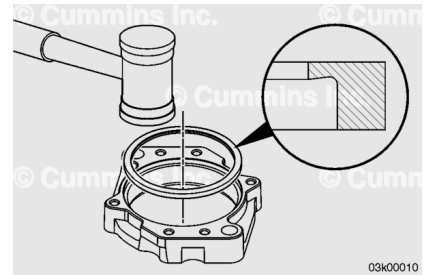


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If the injector cover plate dust seal was damaged and needs to be replaced, lubricate the new dust seal with clean engine oil.

Check that the smooth side of the seal is facing downward and the side with the groove is facing upward.

Use a soft rubber mallet to tap the seal into place.



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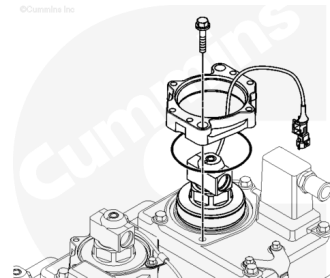
Lubricate the large o-ring and damper on the injector with clean engine oil. Engine oil will protect the o-ring and damper during the installation of the rocker lever cover plate.

Install the new o-ring on the rocker lever cover plate.

Install the rocker lever cover plate on the injector and tap into place with a soft face mallet. Secure to the rocker lever cover plate with two capscrews.

Install the rocker lever cover plate to the rocker lever cover.

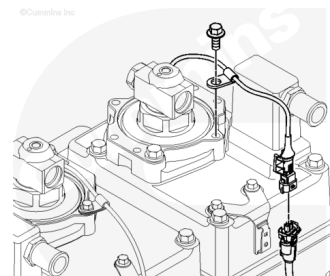
Torque Value: 40 n•m [30 ft-lb]



03400122

Connect the injector wiring harness to the injector.

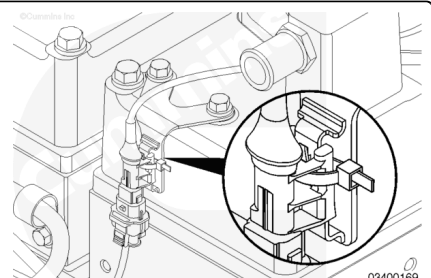
Secure the injector wiring harness to the rocker lever cover with a P-clip.



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Connect the injector wiring to the bracket by sliding onto the bracket until the electrical connection locks into position.

If the locking feature on the electrical connection is missing, use a cable tie, Part Number 3822924, or equivalent, to secure the wiring harness to the bracket.



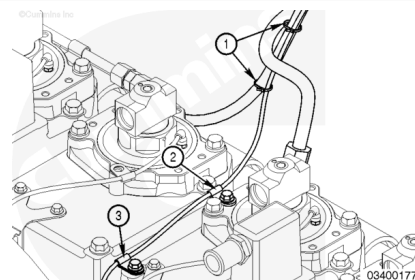
03400169

When installing the rocker lever cover of cylinder number five:

For engines with rear out turbocharger option:

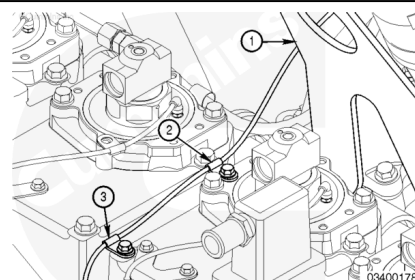
Install two new stainless steel wire ties (1) to secure the turbocharger speed sensor wiring to the turbocharger

Install the P-clips that secure the turbocharger speed sensor wiring to the injector cover plate (2) and to the rocker lever cover (3).



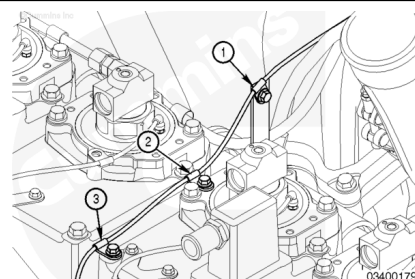
For engines with the front out turbocharger option with a turbocharger elbow bracket:

Install the P-clips (1) that secure the turbocharger speed sensor wiring to the turbocharger elbow bracket, injector cover plate (2), and to the rocker lever cover (3).



For engines with front out turbocharger elbow bracket:

Install the P-clips that secure the turbocharger speed sensor wiring to the "L" bracket (1), injector cover plate (2), and to the rocker lever cover (3).



Finishing Steps

with Mechanically Actuated Injector

- Install air crossover. Refer to Procedure 010-019 in Section 10. (</qs3/pubsys2/xml/en/procedures/20/20-010-019-tr.html>)



with Electronically Actuated Injector

- Install the high-pressure injector supply lines. Refer to Procedure 006-051 in Section 6.
(/qs3/pubsys2/xml/en/procedures/20/20-006-051-tr.html)
- Connect the crankcase breather hose. Refer to Procedure 003-001 in Section 3.
(/qs3/pubsys2/xml/en/procedures/20/20-003-001-tr.html)

